

Gap Analysis – Lab 1 baseline vs. Mini-Project Part-1

Baseline files (kept unchanged in the repository root): Requirements_Table.docx, UseCase_Flow_Specification.docx, UML_UseCase_Diagram.pdf. No source code exists in the repository.

1. EXISTING

Item	Content
Requirements	FR-001...FR-005 and NFR-001, NFR-002 with type, priority, acceptance criteria, rationale (Lab 1 mandated exactly these 5 + 2).
Use case flow	UC-03 Reserve Equipment Slot: preconditions, postconditions, 8-step main flow, alternate flows A1 (calibration) and A2 (slot conflict).
UML use-case diagram	Actors Student, Lab Technician, Notification System; 12 use cases; 3 «include» from Reserve Equipment Slot, 1 «extend» (Apply Late-Return Penalty → Return Equipment).

2. MISSING (required by the Part-1 assignment)

Assignment item	Status before
IEEE-style SRS with introduction, FR, NFR, interfaces	missing → SRS.md
Security objectives (≥ 2) and requirements (≥ 2)	missing → SRS §4 (4 objectives, 6 requirements)
IEEE-style Test Plan, sections 3, 4, 5, §5.1 Security Validation, traceability	missing → Test_Plan.md
≥ 7–10 test cases covering FR and NFR	missing → 23 test cases
Architecture: component diagram + description, pattern, traceability, security architecture	missing → Architecture §2–§7, Figure 2
Design: ≥ 2 sequence diagrams, API design, error handling	missing → Design §8–§11, Figures 3–4
IDs for use cases other than UC-03; use-case descriptions for the rest	missing → UC-01...UC-12 assigned and described

3. NEEDS REVISION

#	Existing material	Problem	Resolution (intent preserved)
1	Diagram vs. FR-004	Diagram links Student to <i>Return Equipment</i> , but FR-004 says a Lab Technician marks the return.	Association removed in the redrawn diagram; documented in SRS §5.1.
2	Diagram vs. requirements	<i>Login, View Reservation History, Update Calibration Status</i> and <i>Send Confirmation Notification</i> have no requirement (NFR-002 mentions calibration update only as a restriction).	Added FR-006, FR-007, FR-008, FR-009 (marked [NEW]).
3	UC-03 precondition "no conflicting reservation for the same time slot"	Not backed by a requirement.	Added FR-010.
4	FR-001	"real-time availability" is not measurable; one sentence bundles viewing, reserving, limits, token.	Kept ID and intent; reworded: availability reflects all Confirmed reservations; limits stated as 120 min / 7 days / overlap; marked [REFINED].

#	Existing material	Problem	Resolution (intent preserved)
5	NFR-001	Bundles latency and lock isolation; "peak load" and "security standards" undefined → untestable as written.	Latency kept as NFR-001 (≤ 200 ms, load = test parameter A-03); isolation outcome split into NFR-003; "security standards" dropped (covered by SEC-REQs).
6	NFR-002	Fine as a requirement but combines RBAC and session expiry.	Kept unchanged; derived SEC-REQ-01 and SEC-REQ-02 so each half has its own test.
7	FR-003	"up to 1 hour before" ambiguous at exactly 60 min.	Assumption A-04 (inclusive), boundary test TC-07.
8	FR-004	Which reservation states can be returned is unstated.	Only Confirmed (Design §9, 409 <code>INVALID_RESERVATION_STATE</code>).
9	UC-03	Secondary actor "Lab Technician" has no step in the main flow; flow ends at A2 with no invalid-session/input, student-overlap or notification-failure paths.	Technician listing kept (fidelity) and footnoted; A3, A4 and E1 added and marked as Part-1 additions.
10	UC-03 step 5/7	The overlap check and the lock are separate steps.	Design performs them atomically in one transaction (DD-02) – same observable behaviour, needed for NFR-003.
11	Diagram	No use-case IDs.	UC-01...UC-12 assigned (UC-03 retained from the Lab 1 specification).

4. Information that Lab 1 does not provide (assumptions A-01...A-12 in SRS §2.6)

Technology stack; definition of "peak load"; slot granularity / minimum duration; behaviour on notification failure; deletion of equipment with future reservations; whether the due date changes status automatically; deployment security (HTTPS). Each was resolved with a labelled assumption – none is stated as fact – and should be confirmed with the instructor.